



CourseShare Hub

A Distributed Course Resource Sharing Platform Built with Django

Course

COMP-8347 Internet Applications and Distributed Systems

Team

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Problem & Motivation

Scattered Resources

Course files live across email, group chats, Brightspace, personal folders, and cloud drives.

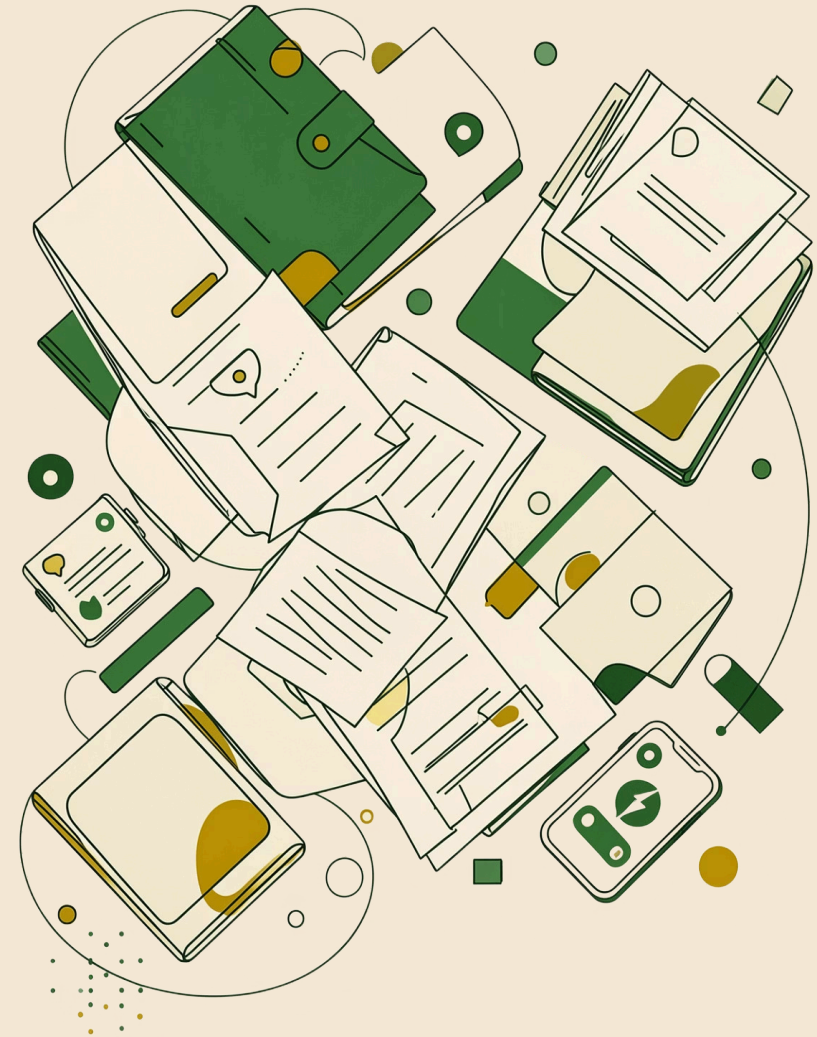
Wasted Time

Students spend too long hunting for notes, lab files, and references when they need them most.

No Shared Hub

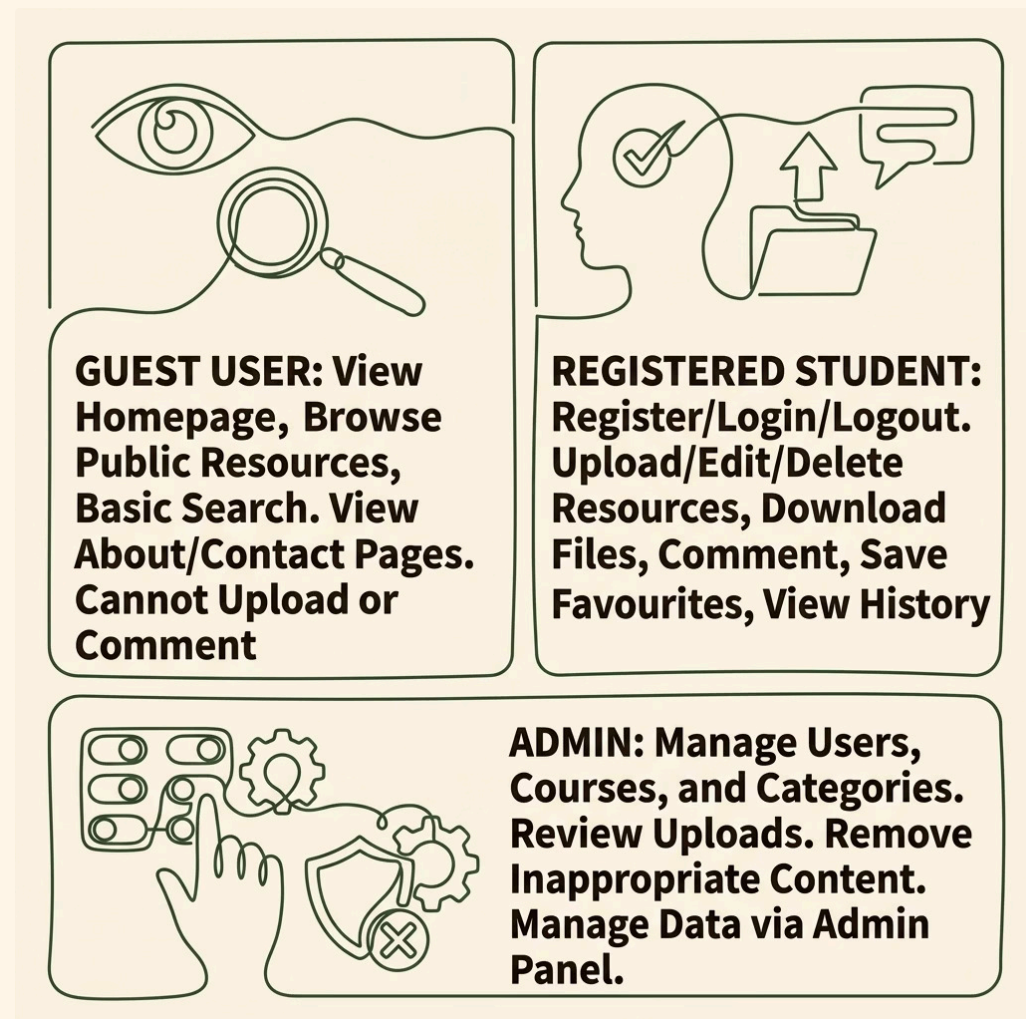
No simple platform exists to upload, categorise, search, and revisit course materials together.

- ✔ CourseShare Hub: one centralised, searchable, student-friendly resource platform.



Project Goal & User Types

Build a structured Django web platform supporting three distinct user roles.



Guest

Browse and search public resources only.

Registered Student

Upload, manage, comment, and save favourites.

Admin

Manage users, categories, and all uploaded content.

Core Features



Authentication

Register, login, logout, forgot password.



Resource Management

Create, view, edit, and delete resources with course and category organisation.



Search & Filter

Keyword search, dropdown filters by course, category, file type, and date.



File Upload

Upload PDFs, documents, and images with title, description, and tags.



Sessions & Cookies

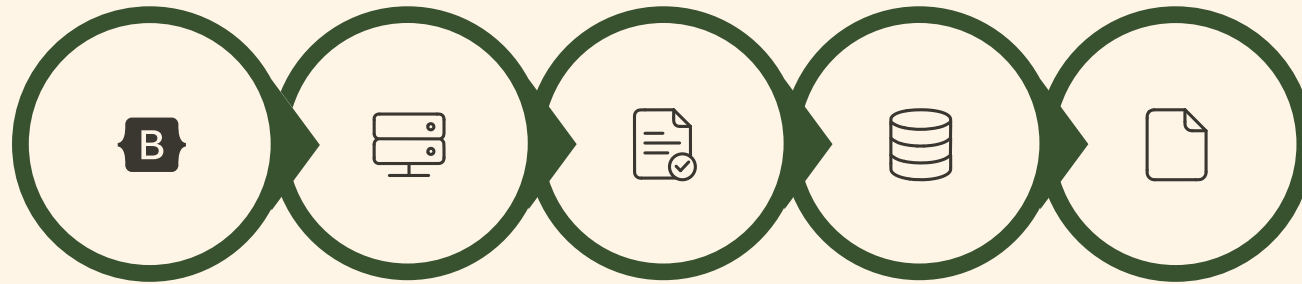
Track visits, recently viewed resources, and recent search keywords.



Dual Interface

Separate experiences for guest and registered users.

Planned Django Implementation



UI: Bootstrap

Django Views

Forms & Auth

Models & DB

JSON Fixtures

Models

Course, Category, Resource,
UserProfile, Comment, Favourite,
UserHistory

Key Tech

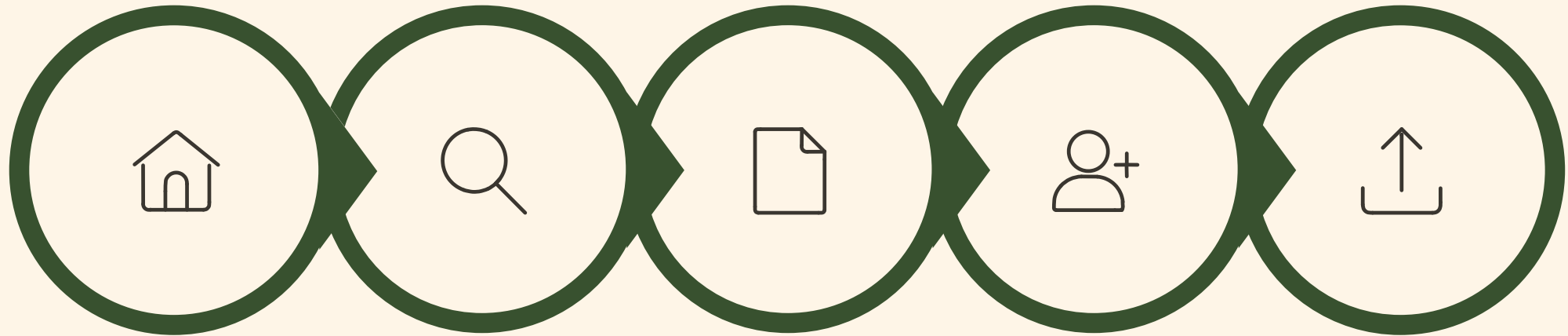
Bootstrap styling · Class-based views ·
JSON fixtures · Sessions & cookies ·
GitHub version control

Group Roles & Responsibilities

Member	Main Role	Key Django Components
Honghao Zhang	Auth & Project Coordination	Registration, login/logout, forgot password, user profile, auth templates
Tianyang Ren	Resource Models & CRUD	Course, Category, Resource models; create/read/update/delete; DB relationships
Lei Jiang	Search, Filter & CBVs	Keyword search, dropdown filters, class-based list and detail views, query logic
Kun Lan	Sessions, Cookies & History	Visit tracking, recently viewed, search history, User History page
Zhihan Zhang	UI/UX, Upload & Testing	Bootstrap layout, file upload interface, About/Contact pages, testing, documentation

 Every member contributes Django code across views, models, forms, and templates.

User Flow & Interface Concept



Homepage

Browse
/Search

Resource Page

Login
/Register

Upload

Homepage Wireframe

Navigation bar · Search bar ·
Login/Register buttons · Course category
cards · Recent resources · Footer

Resource Detail Wireframe

Title · Course · Category · Uploader ·
Download button · Save to favourites ·
Comment section

Upload Page Wireframe

Title field · Course dropdown · Category
dropdown · File upload · Description ·
Submit button

Thank You

CourseShare Hub combines Django's core web development features with a practical academic resource-sharing use case for students.

What

Centralised Django resource sharing platform

Why

Eliminate scattered files, save time, encourage collaboration

How

Models, views, forms, Bootstrap, sessions, file upload, GitHub

COMP-8347 · Group Project Proposal

